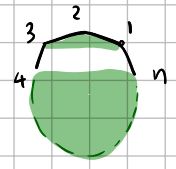
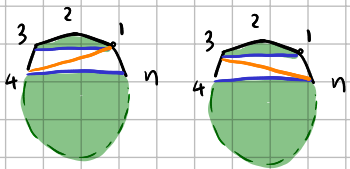
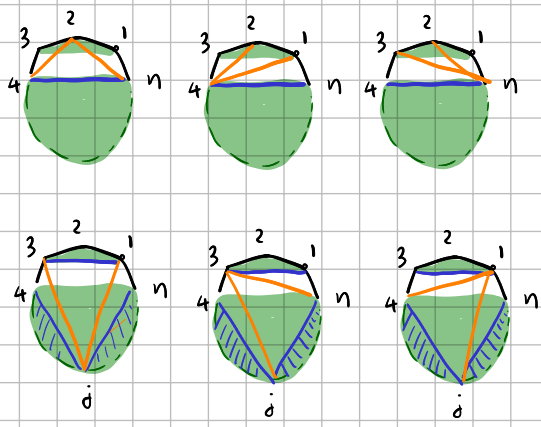


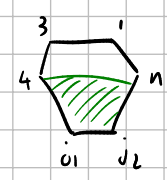
1 BP mit hole $4n$ and 13



2 BP choose either $4n$ or 13

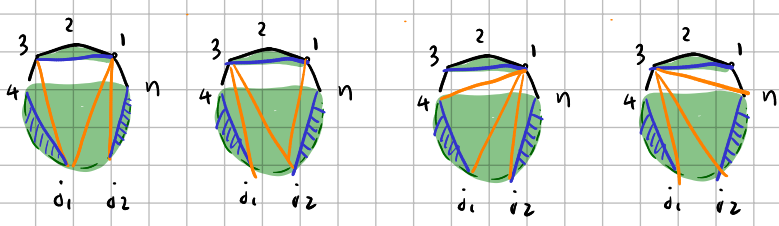


zoom in on hexagon:



→ looks like 1-zero regions

3 BP

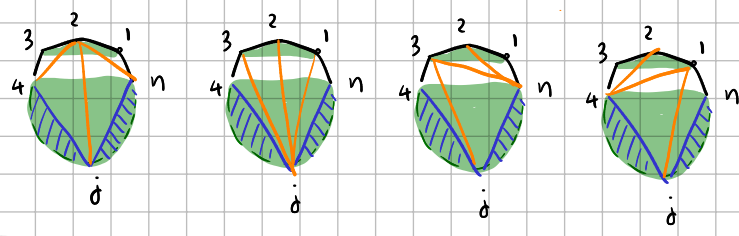


$d_0 = 4$ $d_{541} = n$

$$\sum_{i=0}^{s+1} \frac{1}{\prod_{a=0}^{i-1} X_{10a} \prod_{b=i+1}^s X_{30b}}$$

→ looks exactly like the 1-zero pattern

neither X_{13} nor X_{4n} : z mit omitted

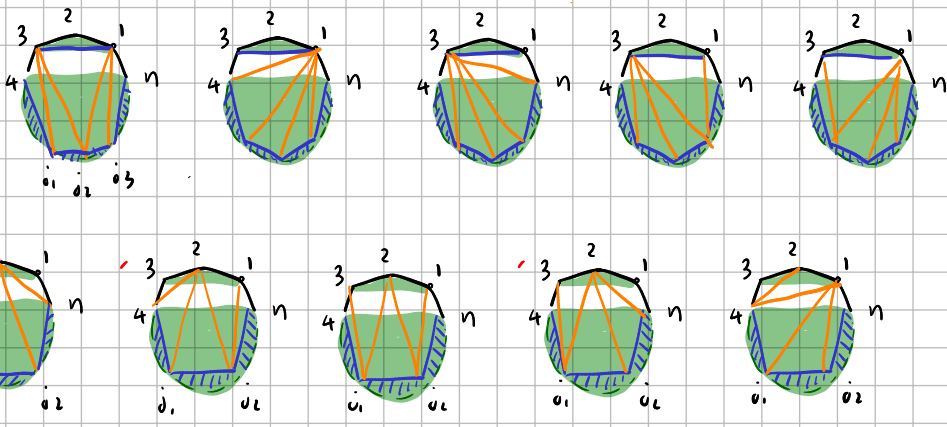


zoom in on hexagon



→ 2-zero hexagon

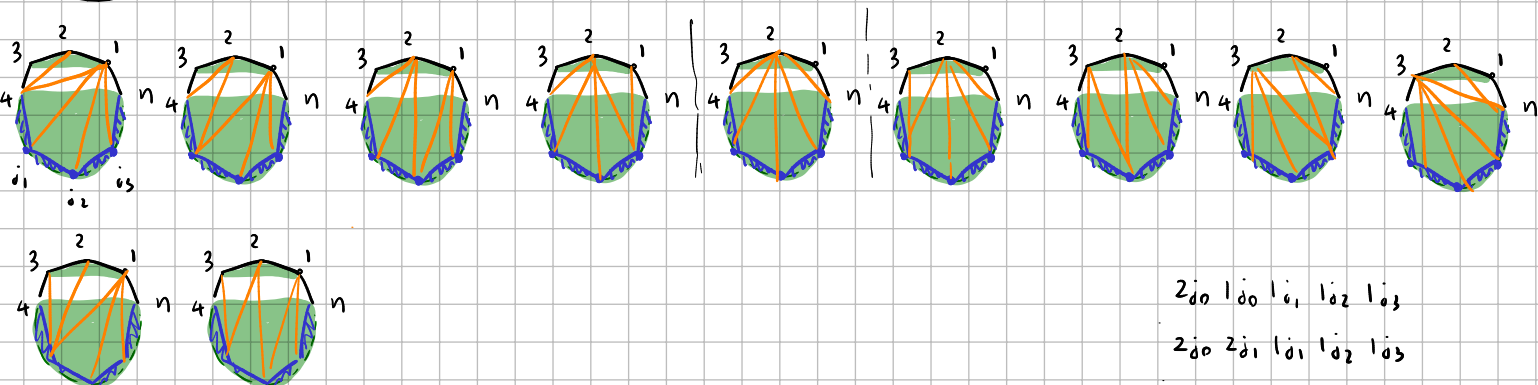
4 BP



$$s=2 \quad d_0=4 \quad d_3=n$$

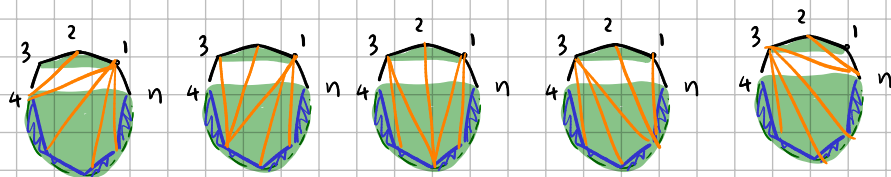
2_{d_0}	2_{d_1}	2_{d_2}	2_{d_3}
2_{d_0}	2_{d_1}	2_{d_2}	1_{d_2}
3_{d_1}	2_{d_1}	2_{d_2}	2_{d_3}
3_{d_1}	2_{d_1}	2_{d_2}	1_{d_2}
2_{d_0}	1_{d_0}	1_{d_1}	1_{d_2}
3_{d_1}	3_{d_2}	3_{d_3}	2_{d_3}

5 BP

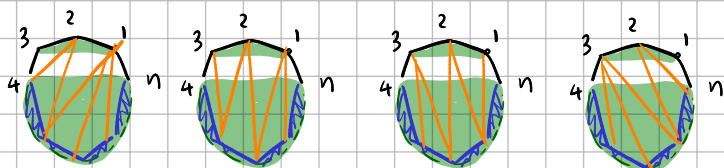


$2_{d_0} 1_{d_0} 1_{d_1} 1_{d_2} 1_{d_3}$
 $2_{d_0} 2_{d_1} 1_{d_1} 1_{d_2} 1_{d_3}$
 $2_{d_0} 2_{d_1} 2_{d_2} 1_{d_2} 1_{d_3}$

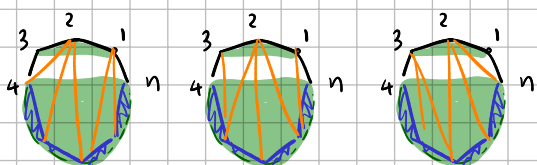
one 2 connected



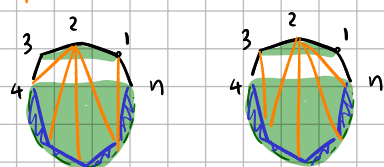
two 2 connected (adjacent)



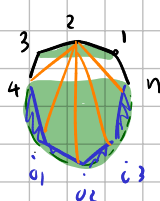
three 2 connected



four 2s



five 2s



sum over $\# 2s$ ($\alpha \equiv \# 2$)

$s+2$: $\#$ of boundary propagators

$\alpha = 1$

$z_{j0}, z_{j1}, \dots, z_{js+1}$

$\alpha = 1$: sum ends at $s+1$

$\alpha = 2$

$z_{j0} z_{j1}, z_{j1} z_{j2} \dots, z_{js} z_{js+1}$

$\alpha = 2$: 1 1 S

$\alpha = 3$

$z_{j0} z_{j1} z_{j2} \dots, z_{js-1} z_{js}, z_{js} z_{js+1}$

$i = 3$: — — — $S-1$

$$\sum_{\alpha=1}^{s+2} \sum_{\ell=0}^{s+2-\alpha} \prod_{l=0}^{\alpha-1} X_{z_{j\alpha+l}} \prod_{b=\alpha-1}^s X_{1;ib} \prod_{c=1}^{\alpha} X_{3;ic}$$

improve:

use α, β index

use α index $\alpha + \ell \rightarrow \alpha$

$\alpha \rightarrow \beta$

$$\sum_{\alpha=1}^{s+2} \sum_{\gamma=0}^{s+2-\alpha} \prod_{\ell=0}^{\alpha-1} X_{z_{j\gamma+\ell}} \prod_{b=\gamma+\alpha-1}^s X_{1;ib} \prod_{c=1}^{\gamma} X_{3;ic}$$

$\alpha = \gamma + \ell$

$\ell = \gamma - \alpha$

α goes from γ to $\gamma + \alpha - 1$

$$\sum_{\alpha=1}^{s+2} \sum_{\gamma=0}^{s+2-\alpha} \prod_{\ell=\gamma}^{\gamma+\alpha-1} X_{z_{j\ell}} \prod_{b=\gamma+\alpha-1}^s X_{1;ib} \prod_{c=1}^{\gamma} X_{3;ic}$$

next stage:

$\gamma + \alpha - 1 = \beta$

so

$\alpha = \beta - \gamma + 1$

(next to keep γ, β as variables)

pairs α, γ :

$$\begin{array}{l} (1, 0), (1, 1), (1, 2) \dots, (1, s+1) \\ (2, 0), (2, 1) \dots (2, s) \\ (3, 0) \dots (3, s-1) \\ \vdots \\ (s+1, 0), (s+1, 1) \rightarrow \gamma=1 \\ (s+2, 0) \end{array}$$

$\gamma=0$ done

β goes from $\gamma, \dots, s+1$

γ goes from 0 to $s+1$

α goes from 1 to $s+2-\gamma$

$\Rightarrow$ that γ is independent and α is dependent

$$DS = \sum_{\alpha=0}^{s+1} \sum_{\beta=\alpha}^{s+1} \left(\prod_{\ell=\alpha}^{\beta} X_{z_{j\ell}} \prod_{b=\beta}^s X_{1;ib} \prod_{c=1}^{\alpha} X_{3;ic} \right)^{-1}$$

α : first index z connects to

β : last index z connects to

prove that 2-zero is sat. sf. id

$$\begin{cases} X_{3ja} = X_{1ja} - X_{1j0} & a = 1, \dots, s \\ X_{3js+1} = -X_{1j0} \\ X_{2js+1} = X_{2j0} - X_{1j0} \\ X_{2ja} = X_{1ja} + X_{2j0} - X_{1j0} & a = 1, \dots, s \end{cases}$$

Common denominator

$$D = \prod_{a=0}^s X_{1ja} \prod_{b=0}^{s+1} X_{2jb} \prod_{c=1}^{s+1} X_{3jc}$$

rename variables

$$x \equiv X_{1u}$$

$$y \equiv X_{2u}$$

$$z_a \equiv X_{1ja} \quad a = 1, \dots, s$$

$$DS \cdot D = \sum_{\alpha=0}^{s+1} \sum_{\beta=\alpha}^{s+1} \frac{1}{\prod_{e=\alpha}^{\beta} X_{2je} \prod_{b=\beta}^s X_{1jb} \prod_{c=1}^{\alpha} X_{3jc}}$$

$$= \sum_{\alpha=0}^{s+1} \sum_{\beta=\alpha}^{s+1} \prod_{b=0}^{\beta-1} X_{1jb} \prod_{c=\alpha+1}^{s+1} X_{3jc} \prod_{a=0}^{\alpha-1} X_{2je} \prod_{e=\beta+1}^{s+1} X_{2je}$$

separate the $j0$ and $js+1$ things

$j0$ only happens when $\alpha > 0$ (so is β)

$js+1$ only happens when $\alpha < s+1$ and $\beta < s+1$

$$= \underbrace{\sum_{\beta=0}^{s+1} \prod_{b=0}^{\beta-1} X_{1jb} \prod_{c=0+1}^{s+1} X_{3jc} \prod_{e=\beta+1}^{s+1} X_{2je}}_{\alpha=0} + X_{1j0} X_{2j0} \underbrace{\sum_{\alpha=1}^{s+1} \sum_{\beta=\alpha}^{s+1} \prod_{b=1}^{\beta-1} X_{1jb} \prod_{c=\alpha+1}^{s+1} X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{e=\beta+1}^{s+1} X_{2je}}_{\alpha > 0}$$

$$= \underbrace{\prod_{c=1}^{s+1} X_{3jc} \prod_{e=1}^{s+1} X_{2je}}_{\alpha, \beta=0} + X_{1j0} \sum_{\beta=1}^{s+1} \prod_{b=1}^{\beta-1} X_{1jb} \prod_{c=1}^{s+1} X_{3jc} \prod_{e=\beta+1}^{s+1} X_{2je} \quad \textcircled{B}$$

$$+ X_{1j0} X_{2j0} \sum_{\alpha=1}^{s+1} \sum_{\beta=\alpha}^{s+1} \prod_{b=1}^{\beta-1} X_{1jb} \prod_{c=\alpha+1}^{s+1} X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{e=\beta+1}^{s+1} X_{2je} \quad \textcircled{C}$$

now extract X_{2jst+1} and X_{3jst+1}

$$(A) \prod_{c=1}^{S+1} X_{3jc} \prod_{a=1}^{S+1} X_{2je} = X_{2n} X_{3n} \prod_{a=1}^S X_{2je} X_{3ja}$$

$$(B) X_{1jo} \sum_{\beta=1}^{S+1} \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=1}^{S+1} X_{3jc} \prod_{a=\beta+1}^{S+1} X_{2je}$$

$$= \underbrace{X_{14} X_{3n} \prod_{b=1}^S X_{1ob} \prod_{c=1}^S X_{3jc}}_{\beta = S+1} + X_{14} X_{2n} X_{3n} \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=1}^S X_{3jc} \prod_{a=\beta+1}^S X_{2je}$$

$\beta < S+1$

$$(C) X_{1jo} X_{2jo} \sum_{\alpha=1}^{S+1} \sum_{\beta=\alpha}^{S+1} \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=\alpha+1}^{S+1} X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{a=\beta+1}^{S+1} X_{2je}$$

$$= \underbrace{X_{14} X_{24} \prod_{b=1}^S X_{1ob} \prod_{a=1}^S X_{2je}}_{\alpha = S+1} + X_{14} X_{24} X_{3n} \sum_{\alpha=1}^S \sum_{\beta=\alpha}^{S+1} \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=\alpha+1}^S X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{a=\beta+1}^{S+1} X_{2je}$$

$$= X_{14} X_{24} \prod_{b=1}^S X_{1ob} \prod_{a=1}^S X_{2je} + \underbrace{X_{14} X_{24} X_{3n} \sum_{\alpha=1}^S \prod_{b=1}^S X_{1ob} \prod_{c=\alpha+1}^S X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je}}_{\beta = S+1}$$

$$+ X_{14} X_{24} X_{3n} X_{2n} \sum_{\alpha=1}^S \sum_{\beta=\alpha}^S \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=\alpha+1}^S X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{a=\beta+1}^S X_{2je}$$

all together

$$X_{2n} X_{3n} \prod_{a=1}^S X_{2je} X_{3ja} + X_{14} X_{3n} \prod_{b=1}^S X_{1ob} \prod_{c=1}^S X_{3jc}$$

$$+ X_{14} X_{2n} X_{3n} \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=1}^S X_{3jc} \prod_{a=\beta+1}^S X_{2je}$$

$$+ X_{14} X_{24} \prod_{b=1}^S X_{1ob} \prod_{a=1}^S X_{2je} + X_{14} X_{24} X_{3n} \sum_{\alpha=1}^S \prod_{b=1}^S X_{1ob} \prod_{c=\alpha+1}^S X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je}$$

$$+ X_{14} X_{24} X_{3n} X_{2n} \sum_{\alpha=1}^S \sum_{\beta=\alpha}^S \prod_{b=1}^{\beta-1} X_{1ob} \prod_{c=\alpha+1}^S X_{3jc} \prod_{a=1}^{\alpha-1} X_{2je} \prod_{a=\beta+1}^S X_{2je}$$

$$\begin{aligned}
 & -(y-x) \times \prod_{a=1}^S (z_a + y - x)(z_a - x) - x^2 \prod_{b=1}^S z_b(z_b - x) \\
 & -x^2(y-x) \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=1}^S (z_b - x) \prod_{a=\beta+1}^S (z_a + y - x) \\
 & + xy \prod_{a=1}^S z_a(z_a + y - x) \\
 & -x^2 y \sum_{\alpha=1}^S \prod_{b=1}^S z_b \prod_{c=\alpha+1}^S (z_c - x) \prod_{a=1}^{\alpha-1} (z_a + y - x) \\
 & -x^2 y(y-x) \sum_{\alpha=1}^S \sum_{\beta=\alpha}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=\alpha+1}^S (z_c - x) \prod_{a=1}^{\alpha-1} (z_a + y - x) \prod_{a=\beta+1}^S (z_a + y - x)
 \end{aligned}$$

$$\begin{cases}
 X_{3\partial a} \rightarrow z_a - x & a = 1, \dots, S \\
 X_{3n} \rightarrow -x \\
 X_{2\partial a} \rightarrow z_a + y - x \\
 X_{2n} \rightarrow y - x & a = 1, \dots, S
 \end{cases}$$

extract z_1 ?

what has z_1 ?

$$z_1^2 + z_1(y - 2x) - x(y + x)$$

$$\begin{aligned}
 \textcircled{1} \quad & -(y-x) \times \prod_{a=1}^S (z_a + y - x)(z_a - x) = -(y-x) \times (z_1 - x)(z_1 + y - x) \prod_{a=2}^S (z_a + y - x)(z_a - x) \\
 & = z_1^2 \left[-(y-x) \times \prod_{a=2}^S (z_a + y - x)(z_a - x) \right] + z_1 \left[-(y-x) \times (y - 2x) \prod_{a=2}^S (z_a + y - x)(z_a - x) \right] \\
 & \quad \left[-(y-x) \times (y - 2x) \prod_{a=2}^S (z_a + y - x)(z_a - x) \right]
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad & -x^2 \prod_{b=1}^S z_b(z_b - x) = -x^2 z_1(z_1 - x) \prod_{b=2}^S z_b(z_b - x) \\
 & = z_1^2 \left[-x^2 \prod_{b=2}^S z_b(z_b - x) \right] + z_1 \left[x^3 \prod_{b=2}^S z_b(z_b - x) \right]
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{4} \quad & xy \prod_{a=1}^S z_a(z_a + y - x) = xy z_1(z_1 + y - x) \prod_{a=2}^S z_a(z_a + y - x) \\
 & = z_1^2 \left[xy \prod_{a=2}^S z_a(z_a + y - x) \right] + z_1 \left[xy(y - x) \prod_{a=2}^S z_a(z_a + y - x) \right]
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{3} \quad & -x^2(y-x) \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=1}^S (z_b - x) \prod_{a=\beta+1}^S (z_a + y - x) \\
 & = -x^2(y-x)(z_1 - x) \prod_{c=2}^S (z_c - x) \prod_{a=2}^S (z_a + y - x) +
 \end{aligned}$$

$$\begin{aligned}
& -x^2(y-x) z_1(z_1-x) \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^S \prod_{a=2}^{\beta+1} (z_b-x) \prod_{a=\beta+1}^S (z_a+y-x) \\
& = z_1^2 \left[-x^2(y-x) \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^S \prod_{a=2}^{\beta+1} (z_b-x) \prod_{a=\beta+1}^S (z_a+y-x) \right] \\
& + z_1 \left[-x^2(y-x) \prod_{a=2}^S (z_a-x) (z_a+y-x) - x^3(y-x) \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^S \prod_{a=2}^{\beta+1} (z_b-x) \prod_{a=\beta+1}^S (z_a+y-x) \right] \\
& + \left[+x^3(y-x) \prod_{a=2}^S (z_a-x) (z_a+y-x) \right]
\end{aligned}$$

$$\begin{aligned}
(5) \quad & -x^2 y \sum_{\alpha=1}^S \prod_{b=1}^S \prod_{c=\alpha+1}^S \prod_{a=1}^{\alpha-1} (z_c-x) \prod_{a=1}^{\alpha-1} (z_a+y-x) \\
& \quad \quad \quad \alpha=1 \\
& = -x^2 y z_1 \prod_{b=2}^S z_b (z_b-x) - x^2 y z_1 (z_1+y-x) \sum_{\alpha=2}^S \prod_{b=2}^S \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c-x) \prod_{a=2}^{\alpha-1} (z_a+y-x) \\
& = z_1^2 \left[-x^2 y \sum_{\alpha=2}^S \prod_{b=2}^S \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c-x) \prod_{a=2}^{\alpha-1} (z_a+y-x) \right] \\
& + z_1 \left[-x^2 y \prod_{b=2}^S z_b (z_b-x) - x^2 y (y-x) \sum_{\alpha=2}^S \prod_{b=2}^S \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c-x) \prod_{a=2}^{\alpha-1} (z_a+y-x) \right]
\end{aligned}$$

$$\begin{aligned}
(6) \quad & -x^2 y(y-x) \sum_{\alpha=1}^S \sum_{\beta=\alpha}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=1}^{\alpha-1} (z_c-x) \prod_{a=1}^{\alpha-1} (z_a+y-x) \prod_{a=\beta+1}^S (z_a+y-x) \\
& \quad \quad \quad \alpha=1 \\
& = -x^2 y(y-x) \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=2}^S \prod_{a=\beta+1}^S (z_c-x) \prod_{a=\beta+1}^S (z_a+y-x) \\
& \quad \quad \quad -x^2 y(y-x) z_1(z_1+y-x) \sum_{\alpha=2}^S \sum_{\beta=\alpha}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c-x) \prod_{a=2}^{\alpha-1} (z_a+y-x) \prod_{a=\beta+1}^S (z_a+y-x) \\
& \quad \quad \quad \beta=1 \\
& = -x^2 y(y-x) \prod_{c=2}^S (z_c-x) (z_c+y-x) - x^2 y(y-x) z_1 \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^S \prod_{a=\beta+1}^S (z_c-x) \prod_{a=\beta+1}^S (z_a+y-x) \\
& \quad \quad \quad -x^2 y(y-x) z_1(z_1+y-x) \sum_{\alpha=2}^S \sum_{\beta=\alpha}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c-x) \prod_{a=2}^{\alpha-1} (z_a+y-x) \prod_{a=\beta+1}^S (z_a+y-x)
\end{aligned}$$

$$= z_1^2 \left[-x^2 y (y-x) \sum_{\alpha=2}^S \sum_{\beta=\alpha}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c - x) \prod_{e=\beta+1}^S (z_e + y - x) \right]$$

$$= z_1 \left[-x^2 y (y-x)^2 \sum_{\alpha=2}^S \sum_{\beta=\alpha}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=2}^{\alpha-1} (z_c - x) \prod_{e=\beta+1}^S (z_e + y - x) \right. \\ \left. -x^2 y (y-x) \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^{\beta-1} (z_c - x) \prod_{e=\beta+1}^S (z_e + y - x) \right] \\ + \left[-x^2 y (y-x) \prod_{c=2}^S (z_c - x) (z_c + y - x) \right]$$

O(1) terms

$$-x^2 y (y-x) \prod_{c=2}^S (z_c - x) (z_c + y - x) + x^3 (y-x) \prod_{a=2}^S (z_a - x) (z_a + y - x)$$

$$-(y-x)x(y-2x) \prod_{a=2}^S (z_a + y - x) (z_a - x)$$

$$\propto [x^3 - x^2 y - x(y-2x)]$$

sanity check: test smallest cases

$$-(y-x)x \prod_{a=1}^S (z_a + y - x) (z_a - x) - x^2 \prod_{b=1}^S z_b (z_b - x) \quad \textcircled{1} \quad \textcircled{2}$$

$$-x^2 (y-x) \sum_{\beta=1}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=1}^{\beta-1} (z_c - x) \prod_{e=\beta+1}^S (z_e + y - x) \quad \textcircled{3}$$

$$+ xy \prod_{e=1}^S z_e (z_e + y - x) \quad \textcircled{4}$$

$$-x^2 y \sum_{\alpha=1}^S \prod_{b=1}^{\alpha-1} z_b \prod_{c=\alpha+1}^S \prod_{a=1}^{\alpha-1} (z_c - x) (z_a + y - x) \quad \textcircled{5}$$

$$-x^2 y (y-x) \sum_{\alpha=1}^S \sum_{\beta=\alpha}^S \prod_{b=1}^{\beta-1} z_b \prod_{c=\alpha+1}^S \prod_{a=1}^{\alpha-1} (z_c - x) \prod_{e=\beta+1}^S (z_e + y - x) \quad \textcircled{6}$$

$$\textcircled{S=1} \quad -(y-x)x(z+y-x)(z-x) - x^2 z(z-x) - x^2 (y-x)(z-x) + xy z(z+y-x) - x^2 y z$$

→ does not work

next, check

$$\sum_{a=0}^{S+1} \sum_{b=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i_{ab}} \prod_{c=\alpha+1}^{S+1} X_{3_{jc}} \prod_{a=0}^{\alpha-1} X_{2_{je}} \prod_{e=\beta+1}^{S+1} X_{2_{je}}$$

→ this is correct

$S=1$:

$$\sum_{a=0}^2 \sum_{b=\alpha}^2 \prod_{b=0}^{\beta-1} X_{i_{ab}} \prod_{c=\alpha+1}^2 X_{3_{jc}} \prod_{a=0}^{\alpha-1} X_{2_{je}} \prod_{e=\beta+1}^2 X_{2_{je}} = 0 \text{ by mathematics}$$

different approach

set $X_{i_{01}}$ to zero to check $O(1)$ term in $X_{i_{01}}$

what does survive?

$$X_{i_{01}} = 0 \Rightarrow X_{3_{01}} = -X_{i_{00}}, X_{2_{01}} = X_{2_{00}} - X_{i_{00}}$$

$$\sum_{a=0}^{S+1} \sum_{b=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i_{ab}} \prod_{c=\alpha+1}^{S+1} X_{3_{jc}} \prod_{a=0}^{\alpha-1} X_{2_{je}} \prod_{e=\beta+1}^{S+1} X_{2_{je}}$$

any time $\beta \geq 2$

we get a factor of $X_{i_{01}}$

only $\beta = 0, 1$ survives

→ only happens when $\alpha = 0, 1$

→ only this survives:

$$\sum_{a=0}^1 \sum_{b=\alpha}^1 \prod_{b=0}^{\beta-1} X_{i_{ab}} \prod_{c=\alpha+1}^{S+1} X_{3_{jc}} \prod_{a=0}^{\alpha-1} X_{2_{je}} \prod_{e=\beta+1}^{S+1} X_{2_{je}}$$

$$= X_{i_{00}} X_{2_{00}} \prod_{c=2}^{S+1} X_{3_{jc}} \prod_{e=2}^{S+1} X_{2_{je}} \quad \alpha=1$$

$$+ \prod_{c=1}^{S+1} X_{3_{jc}} \prod_{e=1}^{S+1} X_{2_{je}} \quad \alpha=0, \beta=0$$

$$+ X_{i_{00}} \prod_{c=1}^{S+1} X_{3_{jc}} \prod_{e=2}^{S+1} X_{2_{je}} \quad \alpha=0, \beta=1$$

$$\propto X_{14} X_{24} + X_{3_{01}} X_{2_{01}} + X_{14} X_{3_{01}}$$

$$= xy - x(y-x) - x^2 = 0 \quad \checkmark$$

There is no $O(1)$ term in $X_{i_{01}}$

$$f(x) = 3x^2 + 2x = 0$$

$$f(n) = 1$$

$$\frac{f(x) - f(n)}{x} \Big|_{x=n} = \frac{3x^2 + 2x}{x} \Big|_{x=n} = 3x + 2 \Big|_{x=n}$$

$$f(x) = 3x^2 + 2x$$

$$g(x) = x$$

$$\frac{f(x)}{g(x)}$$

$$X_{14} = x$$

$$X_{24} = y$$

$$X_{2_{01}} = y - x$$

$$X_{3_{01}} = -x$$

$$X_{3_{01}} = -X_{i_{00}}, X_{2_{01}} = X_{2_{00}} - X_{i_{00}}$$

now divide by $X_{i,0}$ and ob $\lim_{X_{i,j} \rightarrow \infty}$

$$\frac{1}{X_{i,0}} \sum_{\alpha=0}^{S+1} \sum_{\beta=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i,0b} \prod_{c=\alpha+1}^{S+1} X_{j,c} \prod_{a=0}^{\alpha-1} X_{z,ae} \prod_{e=\beta+1}^{S+1} X_{z,je}$$

$$\sum_{\alpha=0}^{S+1} \sum_{\beta=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i,0b} \prod_{c=\alpha+1}^{S+1} X_{j,c} \prod_{a=0}^{\alpha-1} X_{z,ae} \prod_{e=\beta+1}^{S+1} X_{z,je}$$

if we allow $X_{i,0S+1} = X_{i,h}$ as a variable (which we then set to 0)

and $X_{z,00} = X_{z,h}$ as a variable (which we then set to 0)

$$\text{then } \begin{cases} X_{z,je} = X_{i,0e} - X_{i,j0} & e=0, \dots, S+1 \\ X_{z,je} = X_{i,je} + X_{z,j0} - X_{i,0e} & e=0, \dots, S+1 \end{cases}$$

$$\Rightarrow \sum_{\alpha=0}^{S+1} \sum_{\beta=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i,0b} \prod_{c=\alpha+1}^{S+1} (X_{j,c} - X_{i,j0}) \prod_{a=0}^{\alpha-1} (X_{i,je} + X_{z,j0} - X_{i,0e}) \prod_{e=\beta+1}^{S+1} (X_{i,je} + X_{z,j0} - X_{i,0e})$$

only need to split $X_{i,0S+1}$ || problem: $X_{i,0S+1}$ variable is zero

$S+1$ happens when $\beta \leq S$ or $\alpha \leq S$

$$\underline{\alpha = S+1 \quad (\Rightarrow \beta = S+1)}$$

$$\prod_{b=0}^S X_{i,0b} \prod_{a=0}^S (X_{i,je} + X_{z,j0} - X_{i,0e})$$

$$+ \sum_{\alpha=0}^S \sum_{\beta=\alpha}^{S+1} \prod_{b=0}^{\beta-1} X_{i,0b} \prod_{c=\alpha+1}^{S+1} (X_{j,c} - X_{i,j0}) \prod_{a=0}^{\alpha-1} (X_{i,je} + X_{z,j0} - X_{i,0e}) \prod_{e=\beta+1}^{S+1} (X_{i,je} + X_{z,j0} - X_{i,0e})$$

$$= \prod_{b=0}^S X_{i,0b} \prod_{a=0}^S (X_{i,je} + X_{z,j0} - X_{i,0e}) - X_{i,0} \sum_{\alpha=0}^S \prod_{b=0}^S X_{i,0b} \prod_{c=\alpha+1}^S (X_{j,c} - X_{i,j0}) \prod_{a=0}^{\alpha-1} (X_{i,je} + X_{z,j0} - X_{i,0e})$$

$$- X_{i,0} (X_{z,j0} - X_{i,j0}) \sum_{\alpha=0}^S \sum_{\beta=\alpha}^S \prod_{b=0}^{\beta-1} X_{i,0b} \prod_{c=\alpha+1}^S (X_{j,c} - X_{i,j0}) \prod_{a=0}^{\alpha-1} (X_{i,je} + X_{z,j0} - X_{i,0e}) \prod_{e=\beta+1}^S (X_{i,je} + X_{z,j0} - X_{i,0e})$$

chege neres: $X_{1;a} = z_a \quad a = 0, 1, 2, \dots, s$
 $X_{2;0} = y$

$$\rightarrow \sum_{a=0}^s z_a (z_a + y - z_0) - z_0 \sum_{\alpha=0}^s \sum_{b=0}^s \prod_{c=\alpha+1}^s z_b \prod_{a=0}^{\alpha-1} (z_c - z_0) \prod_{a=0}^{\alpha-1} (z_a + y - z_0) +$$

$$- z_0 (y - z_0) \sum_{\alpha=0}^s \sum_{\beta=\alpha}^s \prod_{b=0}^{\beta-1} z_b \prod_{c=\alpha+1}^s (z_b - z_0) \prod_{a=0}^{\alpha-1} (z_a + y - z_0) \prod_{a=\beta+1}^s (z_a + y - z_0)$$

now extract z_1 (by setting it to 0)

$z_1 \rightarrow 0$ only 3rd term with $\alpha \leq 1, \beta \leq 1$ survive.

$$\rightarrow -z_0 (y - z_0) \cancel{(z_1 - z_0)} \cancel{(z_1 + y - z_0)} \prod_{c=2}^s (z_b - z_0) \prod_{a=2}^s (z_a + y - z_0) \quad (\alpha=\beta=0)$$

$$-z_0 (y - z_0) z_0 \cancel{(z_1 - z_0)} \prod_{c=2}^s (z_b - z_0) \prod_{a=2}^s (z_a + y - z_0) \quad (\alpha=0, \beta=1)$$

$$-z_0 (y - z_0) z_0 y \prod_{c=2}^s (z_b - z_0) \prod_{a=2}^s (z_a + y - z_0) \quad (\alpha=\beta=1)$$

$$\propto -z_0 \cancel{(y - z_0)} + z_0 \cancel{(-z_0)} + z_0 y = 0$$

$$\textcircled{A} \sum_{a=0}^s z_a (z_a + y - z_0) = z_0 y z_1 (z_1 + y - z_0) \prod_{a=2}^s z_a (z_a + y - z_0)$$

$$= z_1^2 \left[\prod_{\substack{a=0 \\ a \neq 1}}^s z_a (z_a + y - z_0) \right] + z_1 \left[(y - z_0) \prod_{\substack{a=0 \\ a \neq 1}}^s z_a (z_a + y - z_0) \right]$$

$$\textcircled{B} -z_0 \sum_{\alpha=0}^s \sum_{b=0}^s \prod_{c=\alpha+1}^s z_b \prod_{a=0}^{\alpha-1} (z_c - z_0) \prod_{a=0}^{\alpha-1} (z_a + y - z_0)$$

z_1 appears
 $\alpha \neq 1$

$$= \underbrace{-z_0 y \prod_{b=0}^s z_b \prod_{c=2}^s (z_c - z_0)}_{\alpha=1} - \underbrace{z_0 (z_1 - z_0) \prod_{b=0}^s z_b \prod_{c=2}^s (z_c - z_0)}_{\alpha=0}$$

$$- z_0 y (z_1 + y - z_0) \sum_{\alpha=2}^s \sum_{b=0}^s \prod_{c=\alpha+1}^s z_b \prod_{c=\alpha+1}^s (z_c - z_0)$$

$$= z_1^2 \left[-z_0^2 \prod_{b=2}^S z_b \prod_{c=2}^S (z_c - z_0) - z_0^2 y \sum_{\alpha=2}^S \prod_{b=2}^S z_b \prod_{c=\alpha+1}^S (z_c - z_0) \right]$$

$$+ z_1 \left[-z_0^2 y \prod_{b=2}^S z_b \prod_{c=2}^S (z_c - z_0) + z_0^3 \prod_{b=2}^S z_b \prod_{c=2}^S (z_c - z_0) \right]$$

z_1 appears if : $\beta \geq 2$ or $\alpha \geq 2$ or $\alpha = 0$ or $\beta = 0$

(C)

$$-z_0 (y - z_0) \sum_{\alpha=0}^S \sum_{\beta=\alpha}^S \prod_{b=0}^{\beta-1} z_b \prod_{c=\alpha+1}^S (z_b - z_0) \prod_{a=0}^{\alpha-1} (z_a + y - z_0) \prod_{e=\beta+1}^S (z_e + y - z_0)$$

$\alpha = \beta = 1$

$$= -z_0^2 y (y - z_0) \prod_{c=2}^S (z_b - z_0) \prod_{e=2}^S (z_e + y - z_0)$$

$\alpha = 1, \beta \geq 2$

$$-z_0 (y - z_0) z_0 z_1 y \sum_{\beta=2}^S \prod_{b=2}^{\beta-1} z_b \prod_{c=2}^S (z_b - z_0) \prod_{e=\beta+1}^S (z_e + y - z_0)$$